AAE 451 Request for Proposals (RFP): Fixed-Wing Aircraft Design, Build and Flight Test

Project objectives

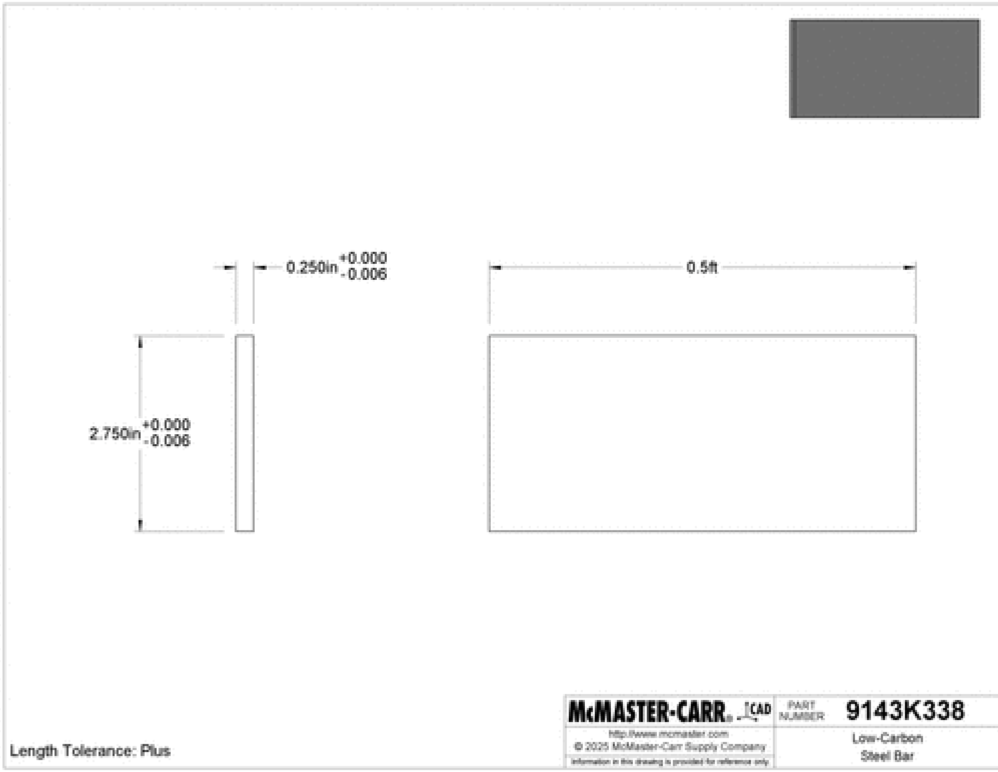
We seek proposals from student teams to design, build, and flight test a remotely piloted, propeller-driven, fixed-winged aircraft. The primary objective is to create an aircraft that can take-off from the runway, safely fly three laps at McAllister Park airfield in Lafayette, IN (as shown below, takeoff direction is subject to wind), and perform a soft field landing.

Project requirements

The project requirements, other than those mentioned above, are the following:

- The aircraft must be stored within a 75 cm width x 75 cm height x 150 cm length storage volume when not actively being flown or worked on.
- The aircraft wingspan can be up to 150 cm.
- The aircraft Maximum Takeoff Weight (MTOW) cannot exceed 58.84 N (mass = 6 kg).
- The forward most ground contact point of the aircraft must begin on the start line of the paved runway. The aircraft must take-off (all ground contact points off the ground) within 25 meters of the start line.
- After takeoff, the aircraft will climb to an altitude of 30 m.
- The aircraft must be capable of performing a soft field landing (packed dirt base, potentially wet, with grass up to 10 cm in height) without sustaining damage.
- The flight operation must not disturb or damage local wildlife, bystanders, vegetation, or private property.
- The complete propulsion system must not consume more than 1000 W of electrical power.
- The battery must be safely connected (without any appendages crossing the arc of the propeller) without removing any aircraft components.
- The aircraft must demonstrate good flight test attributes:
  - Transition from the storage to flight configuration (minus battery and payload) in less than 5 minutes.
  - The payload and battery must be installed and secured in less than 2 minutes.
  - Stable and controllable with and without payload.

    - Easy to fly by a pilot external to the team.
- The aircraft must carry a minimum payload of 1 steel bar shown below, mass = 518 g +/- 1 g.



Provided avionics and data collection

Each team will be provided with the following control, avionic, and sensor components:

- 2.4 GHz transmitter
- 2.4 GHz receiver
- Pixhawk 6C mini flight controller and included cables
- Power module for Pixhawk
- GPS antenna
- Digital airspeed sensor

Teams will be responsible for incorporating the provided electronics into their aircraft design and utilizing them for aircraft control and collecting the following parameters during flight testing:

- GPS altitude, groundspeed, climb rate, and distance flown
- Airspeed
- Aircraft acceleration, load factor, turn rate, and roll rate

Materials and budget

In addition to the provided electronics, each team will have a budget of $400 to purchase materials and items specific to their aircraft. Materials and components sourced from the lab, such as hardware, 3D printed parts, motors, ESCs, propellers, control linkages, and servos, shall be included in the budget. Components manufactured using University resources (i.e. build lab or BIDC equipment) must only include the material cost and not the machine time cost (outsourced parts must include machine/labor cost as well as material cost). Teams must order off-the-shelf components from a list of approved suppliers. Any deviations must be formally requested in writing to the customer (the teaching team) and justified using sound engineering and business logic.

Evaluation

Aircraft will be evaluated on their ability to complete the outlined mission while meeting all specified requirements.

Timeline and deliverables

Flight readiness reviews will be conducted in Week 11 of the semester. Initial flight tests will be held in Week 12 with additional flight tests in Week 13 if needed. Final presentations will be held during Week 15. Poster session will be held Friday, December 12 from 1pm - 3pm in the ARMS Atrium.